

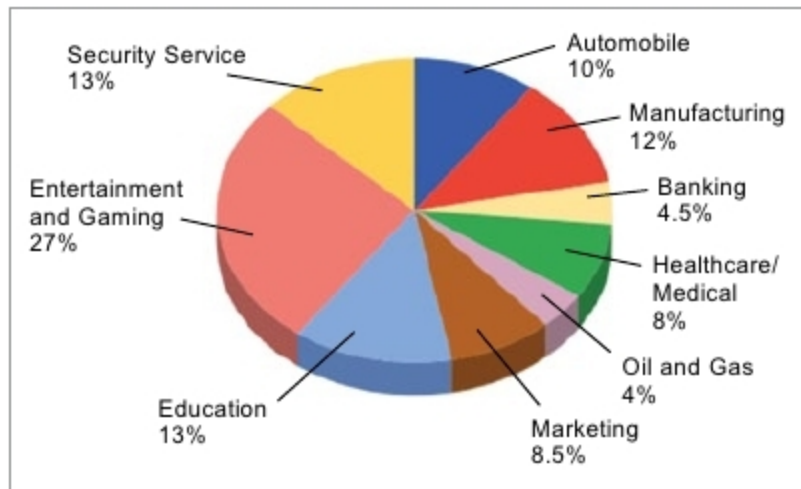


Fig. 1: VR and AR application field and their market share

as reality. This technology is bringing scores of innovative opportunities for upcoming and existing businesses and trade. Through AR, people from various professions are being benefitted and can impose the graphics generated from computers in their field-of-vision (FOV).

In contrast to AR,

virtual reality (VR) can be experienced through devices that make people fully immersed in simulated or non-realistic environments. The basic difference between the two is that VR produces an artificial environment, whereas AR simply uses the current environment by superimposing a new environment or objects on top of it. Specially-designed headsets and glasses can be used for visual immersion, while handhelds and wearables offer tactile immersion.

Currently, both technologies have gained massive demand and popularity across such applications as healthcare, life and sciences, media, gas and oil, tourism, education and training, gaming, entertainment, military, marketing, e-commerce and retail. But market researchers have now predicted that retail and e-commerce will need to push this technology more because of escalating competition among industry tycoons.

Sudden augmentation of demand for mobile devices and efflux in R&D investments are the most imperative factors responsible for the growth of the worldwide AR and VR market.

The healthcare sector is also using this technology for training through 3D modelling. As per market insights provided by MediaTek, Imagine and Institution of Engineering and Technology India, different applications of AR and VR share an approximate market share as represented in the 3D pie-chart given in Fig. 1.

Companies investing in AR/VR

Unlike other consumer electronics available in the market, demand and adoption ratio of AR/VR technology is a quite low, which is also due to the high price and impact of VR headsets on eyes. Many of the leading multinational tech companies are now investing and preparing budgets for VR/AR R&D so that the technology progresses more.

Microsoft is now said to be leading in this technology, as along with millions in investment it also owns the rights for over 10,000 AR/VR patents across thousands of patent families. Apart from Microsoft, Intel and Sony are also investing a lot in this technology for its enhancement.

It is not only buyers and media

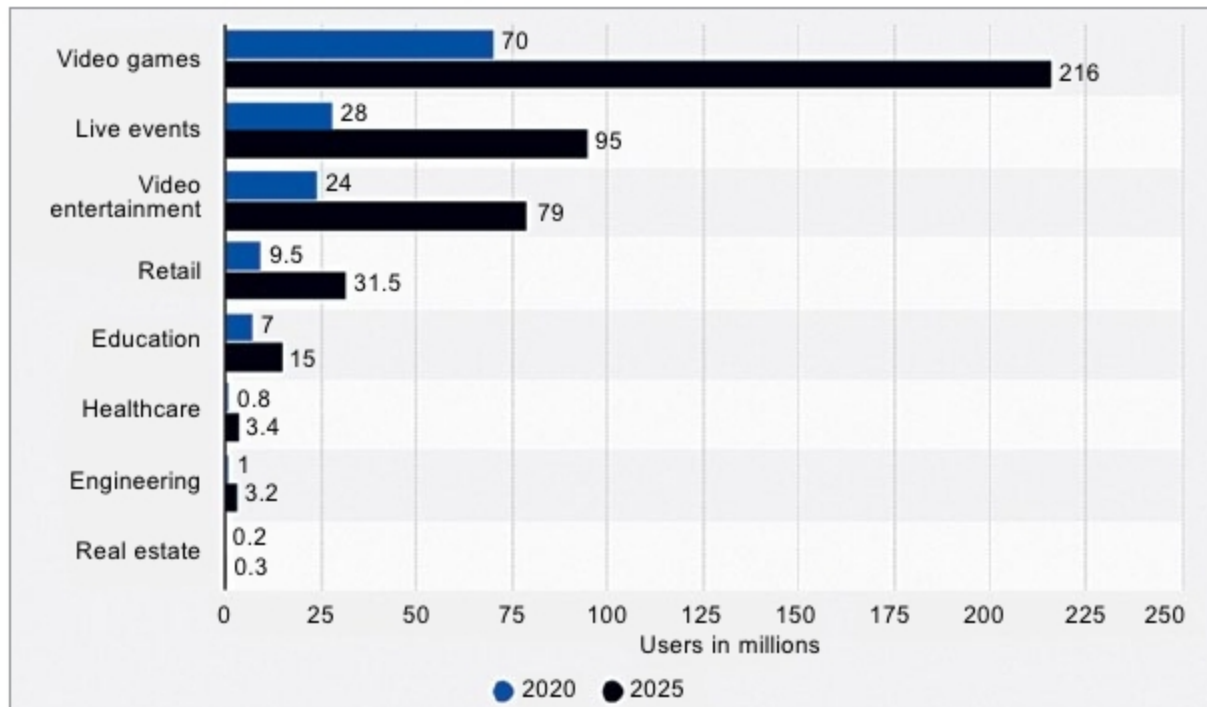


Fig. 2: Forecast user base of the augmented and virtual reality software market worldwide in 2020 and 2025, by segment (in millions) (Source: Goldman Sachs Statista 2018)

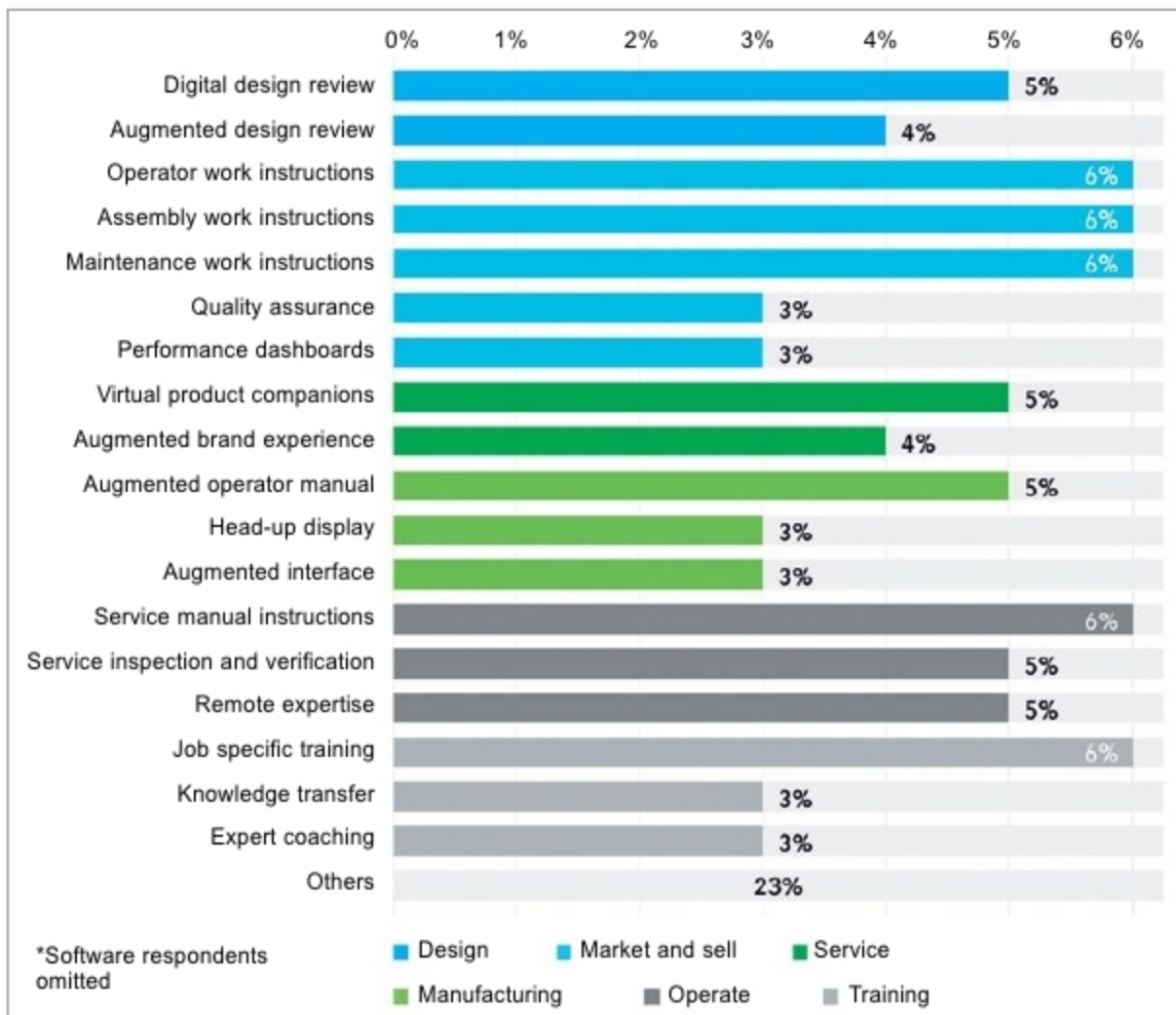


Fig. 3: AR use-case adoption (Source: ptc.com—The State of Industrial Augmented Reality 2019)